

Pulte Metrics — Audit Reconciliation

External metric-vs-requirement findings vs. corrections applied in jtbd_pulte_Corrected.xlsx
Prepared July 7, 2026

Summary

This document maps an independent metric-vs-requirement audit against the corrections already applied. Of ~28 findings: roughly **16 fully fixed**, **~2 partial**, **~7 open**, plus 2 nuances. The applied corrections targeted the calculation contradictions that produced wrong dollar figures on the Key Detail screens (retirement 60%, reserves, required-to-close, DTI). Most of the remaining items are scope realignment (metrics living under the wrong JTBD) and logic/banding mismatches.

Legend: **FIXED** = corrected in file. **PARTIAL** = improved but not fully closed. **OPEN** = not yet changed. **NUANCE** = judgment call / kept intentionally.

Clear contradictions

Metric / item	Status	What that means now
AST-TYPE Total Eligible Assets	FIXED	Retirement changed from a 60% haircut to 100% per Terms & Conditions of Withdrawal.
AST-RSLE Reserves (Months)	FIXED	Now Assets-After-Closing / PITIA, net of cash-to-close and account minimums; no 60% haircut.
AST-FCDP Required to Close	FIXED	Now includes prepaids, debt payoff, EMD, gift, grants/DPA, 1031, sale proceeds; compares Total Eligible Assets (not current_balance).
AST-GIFT Gift Funds	FIXED	Indicator de-scoped to gift-letter presence + amount; donor capacity / proof of transfer / source routed to AST-GDNR. (Group A applied.)
APR-BASE Property Condition	FIXED	Re-banded: C1-C4 eligible (C4 subject to holdback); C5-C6 ineligible until cured. (Group A applied.)
APR-BASE Property Quality	FIXED	Re-banded: Q1-Q5 eligible; Q6 ineligible. (Group A applied.)
IDN-DPV ID Expiration	FIXED	Expired ID now valid with a documented extension; otherwise unexpired through closing. (Group A applied.)
INS-HO Policy Effective	FIXED	Now tests effective_date <= closing date and in force through closing (end_date beyond closing); no longer uses today. (Group A applied.)
TAX-CAL Escrow for Taxes	FIXED	Bug fixed: is_escrow_waived=true now resolves to 'Waived' (required-but-waived flags 'Waived - Discrepancy'). (Group A applied.)
TTL-ELG Chain of Title	FIXED	Now applies FHA 90-day flip seasoning (91-180 day second-appraisal tier); conventional continuity-of-obligation. (Group A applied.)

Unsupported / wrong-scope metrics

All still **OPEN** — no metric was moved to a different JTBD in this pass:

Metric / item	Status	What that means now
APR-META Appraisal LTV	OPEN	Requirement does not define appraisal LTV / max-LTV comparison.
APR-14U	OPEN	Metrics display appraiser/GLA/beds/baths; requirement focuses on waiver gate, Form 1004, completeness, PUD HOA dues.
AST-DATP Down Payment	OPEN	Requirement is statement completeness/authenticity/date coverage; metric is only down-payment amount.
CRD-CPIR (grading/collections/AU/utilization)	OPEN	Several metrics not supported by that requirement block. Payment-history severity and collections treatment were improved in wording, but the scope placement remains.
INC-VER income/trend metrics	NUANCE	Audit is right that INC-VER is a verification JTBD. Rather than relocate, the existing metric was made accurate (calculated, not disclosed). Relocation is a larger restructure.
INC-WAG Total Monthly Debt	OPEN	Debt summation belongs under Liabilities; duplicate of LIA-BAS.
INS-HO Mortgage Insurance premium	OPEN	Out of scope for homeowner's insurance declaration review.
TTL-COV Liens	OPEN	Requirement is title coverage/endorsements/exceptions; metric classifies liens from liability records.

Too narrow / missing logic

Metric / item	Status	What that means now
CRD-DCAM Credit Report Date	NUANCE	Fixed the date basis (note/closing date vs 'today'). The 120-day rule is agency-standard (FNMA B1-1-03) and was kept; its absence from the requirement block is a requirement gap, not a metric error.
CRD-CPIR Recent Inquiries	FIXED	Now counts only 90-day inquiries without a matching new tradeline. (Group A applied.)
INC-WAG VOE Status	FIXED	Per SME decision, redefined as 'Written VOE': reports Received / Not Received when a written VOE (Form 1005) OR a third-party vendor form (The Work Number/Equifax, Truv, Experian Verify, Finicity) is present. Verbal VOE, paystub, W-2, and bank statements no longer count. Timing / per-employer enforcement intentionally not included per SME.
LIA-PMT Revolving/Installment	PARTIAL	Added guidance for \$0-revolving 5% fallback, student-loan, 10-month exclusion, leases; still relies on upstream data honoring it.
LIA-DTI Front/Back DTI	PARTIAL	Back-end now PITIA-inclusive; front-end reconciled to one value; adjusted-DTI workflow not added.

Metric / item	Status	What that means now
LPE-DTMC Score/DTI thresholds	OPEN	Omits Pulte overlay conditions, LTV/occupancy conditions, AUS/exception paths, credit depth. Overlay encoding still pending.
LPE-AMLV LTV	FIXED	Added CLTV (LPE_AMLV_LM4) and HCLTV (LPE_AMLV_LM5) metrics. (Group A applied.)
TAX-CAL Annual Property Tax	FIXED	Added source hierarchy (cert>bill>title) + new-construction reassessment; recurring-only summation. (Group A applied.)
TTL-AUS Lender Policy	FIXED	Added coverage test: Sufficient when coverage >= original principal loan amount; else Insufficient. (Group A applied.)

No issues noted

Consistent with the audit, no material metric-vs-requirement issues stand out for IDN-NUS, LPE-GLPC, or TTL-BAS.

Remediation plan — currently in the metric vs. what the fix should be

Each row shows the current (as-is) metric text and the proposed change, plus where to apply it. Group A is already APPLIED; B/C/D are pending.

A. Metric fixes — single agency-correct answer (APPLIED)

Item	Fix in	Currently in the metric (as-is)	What the fix should be	Where
TAX-CAL Escrow	Metric	Both branches output 'Required' — is_escrow_waived=true still returns 'Required' (copy-paste bug).	is_escrow_waived=true -> 'Waived'; required-but-waived -> 'Waived - Discrepancy'.	TAX_CAL_LM3 indicator
APR Property Condition	Metric	'Good' when C1-C3; 'With Issues' when C4-C6.	'Eligible' C1-C4 (C4 = holdback); 'With Issues' C5-C6 (ineligible until cured).	APR_BASE_LM1 indicator
APR Property Quality	Metric	'Good' when Q1-Q3; 'With Issues' when Q4-Q6.	'Eligible' Q1-Q5; 'Ineligible' Q6.	APR_BASE_LM2 indicator
IDN ID Expiration	Metric	Days to earliest ID expiration vs closing; Valid/Expiring/Expired bands. No allowance for a documented extension.	Expired ID valid WITH a documented extension; else unexpired through closing.	IDN_DPV_LM10/11/12 indicator
INS-HO Policy Effective	Metric	'Active' when effective_date <= TODAY; 'Not Yet Effective' when > today.	Test effective_date <= closing/first-payment date AND in force through closing; drop 'today'.	INS_HO_LM4 indicator
CRD Recent Inquiries	Metric	Counts ALL hard inquiries within 90 days.	Count only 90-day inquiries WITHOUT a matching new tradeline.	CRD_CPIR_LM4/5/6 indicator
TTL-ELG Chain of Title	Metric	'Clear' when unbroken 24+ months, no flip; related-party within 6 months flagged. No FHA 90-day rule.	FHA 90-day flip seasoning (91-180 second-appraisal tier); conventional continuity-of-obligation.	TTL_ELG_LM1 indicator
TTL-AUS Lender Policy	Metric	Shows coverage amount; indicator Received/Ordered/Not Ordered only.	Add coverage test: Sufficient when coverage >= original principal loan amount, else Insufficient.	TTL_AUS_LM3 indicator
AST-GIFT Gift Funds	Metric	Indicator = 'Documented' only if gift letter + proof of transfer + donor ability ALL present (over-scoped).	Gate on gift-letter + amount only; route proof-of-transfer/donor ability/source to AST-GDNR.	AST_GIFT_LM1/2/3 indicator
LPE-AMLV CLTV/HCLTV	Metric (add)	No CLTV/HCLTV metric exists — only LTV (LPE_AMLV_LM2).	Add CLTV and HCLTV metrics (subordinate liens; full HELOC line for HCLTV).	New LPE_AMLV_LM4/LM5
TAX Annual Property Tax	Metric	Lists recurring components; excludes non-recurring. No source hierarchy or new-construction reassessment.	Add source hierarchy (cert>bill>title) + new-construction reassessment (land+improvements).	TAX_CAL_LM1 indicator/calc

B. Metric fixes needing SME input or platform (PENDING)

Item	Fix in	Currently in the metric (as-is)	What the fix should be	Where
LPE-DTMC (Pulte overlays)	Metric + SME	'Compare to min credit score from JTBD validation rules' (generic; no encoded floors).	Encode score floors by occupancy+LTV (primary >80%=600; 2nd/inv >80%=620; <=80% none) + LTV/occupancy/AUS/depth.	CRD_DCAM_LM4-6 + LPE_DTMC_LM1
LIA-PMT imputed payments	Metric + platform	Sums monthly_payment_usd as-is (no imputed payments).	Impute: \$0 revolving = greater of \$10/5%; student 1%/0.5%/IDR; <=10-mo excluded; leases counted; HELOC on drawn.	LIA_PMT_LM1-6 calc + data field
LIA-DTI adjusted DTI	Metric	Back-End DTI = 'Total Monthly Debt Payments / income' (ambiguous re PITIA; no adjusted DTI).	Make PITIA-inclusive: (Total Monthly Debt + PITIA)/income; add an adjusted/total DTI.	LIA_DTI_LM2 + new metric

C. Guideline (JTBD) fixes — metric is correct, requirement is the gap (PENDING)

Item	Fix in	Currently in the metric (as-is)	What the fix should be	Where
CRD-DCAM 120-day	Guideline JTBD	Metric measures 120-day currency (correct); the requirement doesn't state the 120-day rule.	Add the 120-day credit-validity rule to the CRD-DCAM requirement.	CRD-DCAM requirement
CRD-CPIR extra metrics	Guideline JTBD	Metrics exist (payment history/collections/AU/utilization) but the requirement doesn't authorize them.	Expand the requirement to authorize them, or relocate to a credit-analysis JTBD.	CRD-CPIR requirement
INC-VER income/trend	Guideline JTBD	Metric computes total income/trend though the requirement says INC-VER only verifies documentation.	Update the requirement to acknowledge INC-VER produces income, or relocate the calc metrics.	INC-VER requirement
APR-META Appraisal LTV	Guideline JTBD	Metric computes appraisal LTV vs max, but the requirement doesn't define it.	Add appraisal-LTV/max-LTV language to APR-META, or move the LTV metric to LPE-AMLV.	APR-META requirement

D. Scope / relocation (PENDING)

Item	Fix in	Currently in the metric (as-is)	What the fix should be	Where
INC-WAG Total Monthly Debt	Metric (relocate)	Duplicate Total Monthly Debt metric under INC-WAG (also under LIA-BAS).	Remove the duplicate; keep the single source under LIA-BAS.	Delete INC_WAG_LM10/11/12
AST-DATP Down Payment	Metric (relocate)	Metric = down-payment amount only; the JTBD is statement completeness/authenticity/date.	Move Down Payment to AST-FCDP; add AST-DATP completeness/date metrics.	AST_DATP_LM1 -> AST-FCDP
INS-HO MI premium	Metric (relocate)	MI premium handled under homeowner's-insurance review (out of scope).	Move MI premium to the MI JTBD / PITIA build; keep INS-HO for hazard.	-> INS-MI-REQ/INS-PMI-ELG
TTL-COV Liens	Metric (re-source)	Liens counted from Liability records; the JTBD is title coverage/endorsements/exceptions.	Re-source liens from the title commitment (Schedule B exceptions).	TTL_COV_LM1 data source
APR-14U	Metric (re-scope)	Metrics display appraiser/GLA/beds/baths; the JTBD focuses on waiver gate/1004/PUD dues.	Add waiver-gate/Form 1004 completeness/PUD HOA-dues metrics; keep GLA/beds/baths as display.	APR_14U_LM1-4

Recommended next batch (deterministic, ready to apply)

1. TAX-CAL escrow bug — fix the second branch so escrow-waived resolves to 'Waived' (not 'Required').
2. AST-GIFT indicator — de-scope to gift-letter presence + amount; route donor capacity / proof of transfer / source to AST-GDNR.
3. APR-BASE — Property Condition: C1-C4 eligible (C4 holdback); Property Quality: Q1-Q5 eligible, Q6 ineligible.
4. IDN-DPV — treat expired IDs with an acceptable documented extension as valid.
5. INS-HO Policy Effective — test effective_date <= closing date and policy in force through closing (not <= today).
6. CRD-CPIR Recent Inquiries — count only inquiries without a matching new tradeline.
7. TTL-ELG — align flip seasoning to 90 days; TTL-AUS — require lender coverage >= original principal loan amount.

Separate, larger pass: scope realignment (move metrics to the correct JTBD) and LPE-DTMC Pulte-overlay encoding.